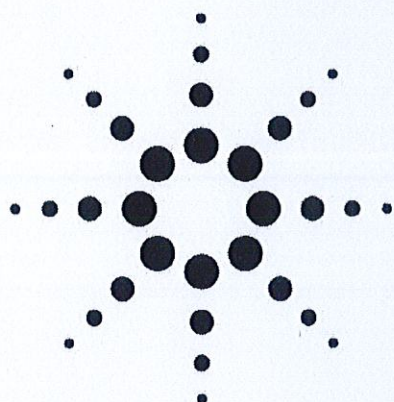




SYSTEM OPERATIONAL TEST

Date Verification

DD: 27	MM: 05	YYYY: 2026
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Instrument Detail

Instrument ID	7820A		
Model No	G840B		
Serial No	CN16302004		
Signature			
Name	LAM HUYNH KHANG		



INSTRUMENT: GC (Gas Chromatography)

☐ 6890 ☐ 7890 ☒ 7820 ☐

General System :

Front Inlet

☐ Split/Splitless
☐ Cool On Column
☐ PTV
☐ Purged Packed
☒ Not Install

Back Inlet

☒ Split/Splitless
☐ Cool on Column
☐ PTV
☐ Purged Packed
☐ Not Install

MSD

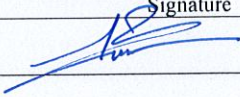
☐ 5973 ☐ 5973N ☐ 5975 ☐ 5977
☐ 7000B/C/D ☒ Not Install
☐ EI ☐ CI

Front Detector

☐ FID ☐ FPD
☐ TCD
☐ uECD
☐ NPD
☒ Not Install

Back Detector

☒ FID ☐ FPD
☐ TCD
☐ uECD
☐ NPD
☐ Not Install

Customer Details		
Customer Name (Print)	Signature	Date
CÔNG TY CỔ PHẦN ĐỒNG XANH		
Qualified Service Representative's Name	Signature	Date
LÂM HUỖNH KHANG		27/05/2026
Company Name	Location	
CÔNG TY TNHH VINTEK	280/130 Bui Huu Nghia, Gia Dinh, TP HCM	
Department :	Contact : ☐ 1 time / year ☒ 2 times per year Frequency of OQ/PVs	

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

Date Printed: 09-Jun-2023 15:51:34

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EQUIPMENT LIST

Equipment	
Ohmmeter	Measure continuity of AC ground (calibration not required)
Digital flowmeter	At least 1.0-100 ml/minute +3% (calibrated)
Digital mass flowmeter	0-5 ml/minute: accuracy $\pm 5\%$, with a minimum of 0.2 ml/minute 5-500 ml/minute: accuracy $\pm 2\%$ (calibrated)
Digital thermometer	Display capable to 1.0°C (calibrated)
Temperature probe	Measure at 100°C and 230°C $\pm 2^\circ\text{C}$ (calibrated)
Digital manometer	Capable of 100 psi at 50% FS, accuracy 0.2% (calibrated)
Samples	
FID Standard Kit (18710-60170)	0.033% (w/w) of C ₁₄ , C ₁₅ , C ₁₆ normal paraffin hydrocarbons in hexane
FPD Standard Kit (19305-60580) (Old style FPD)	20 ng/ μl each of dodecanethiol and tributylphosphate in iso-octane
FPD Standard Kit (5188-5953) (New style FPD)	2.5ng/ μl Methyl Parathion in iso-octane
ECD Standard Kit (18713-60040)	0.033 ng/ μl lindane and Aldrin in iso-octane
ECD Standard Kit Recommended (5183-0379)	0.033 ng/ μl lindane in iso-octane
NPD Standard Kit (18789-60060)	0.65 ng/ μl azobenzene, 1000 ng/ μl octadecane and 1.0 ng/ μl malathion in iso-octane
Headspace Standard Kit (5182-9733)	0.2% v/v each of di-chlorobenzene, nitrobenzene and t-dibutylsulfide in ethanol (includes required micropipettes)
Response Linearity Standard Kit (5188-5244)	1.5 ng/ μl of C ₁₇ , 10 ng/ μl of C ₁₈ , 2 ng/ μl of C ₁₉ , 100 ng/ μl of C ₂₀ , 1000 ng/ μl of C ₂₂ , and 10000 ng/ μl of C ₂₄ normal paraffin hydrocarbons in iso-octane
Columns	
5 m x 0.1 mm retention gap	19091-60620(E) or equivalent (SS/FID (excl. response linearity test), SS/TCD, HS/SS/FID)
HP-5 30 m x 0.32 mm x 0.25 μm	19091J-413(E) or equivalent

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

Date Printed: 09-Jun-2023 15:51:34

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SYSTEM TEST

Basic Safety and Operation Test	Safety Ground Confirmation	
	1. Use the power switch to turn off the GC and then unplug the power cord	
	2. Unplug the network connection from the GC	
	3. Use an Ohmmeter to measure continuity between the GC chassis and the electrical plug's ground pin and record the result.	
	Result	☒ Passed ☐ Failed
Note		
	Safety Ground Confirmation	
	1. Make sure that the GC is not in active use	
	2. Power Cycle the instrument, observe the self-test and record the result	
	Result	☒ Passed ☐ Failed
	Request an oven temperature of 350°C and record the result	
	Result	☒ Passed ☐ Failed
Note		
	Oven Shutdown	
	With the oven on and at 50°C, open the door and record the result.	
	Result	☒ Passed ☐ Failed
Note		
	Hydrogen Safety Shutdown	
	1. Configure any installed inlet for hydrogen carrier gas and, if present, disconnect the column or inlet plug.	
	2. Request an inlet pressure of 100 psi and record the result	
	3. Power cycle the instrument to recover from the error and then restore the inlet's original carrier gas configuration.	
	Result	☒ Passed ☐ Failed
Note		

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

Date Printed: 09-Jun-2023 15:51:34

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Oven Temperature Accuracy Test		
Oven Temperature Accuracy Test	Set point of 100°C	
	1. Set the oven temperature to 100°C	
	2. Turn on the oven wait the oven at least 5 minutes after the oven is ready	
	3. Observe the temperature on the calibrated thermometer's display and ,when the temperature stabilizer, record the reading	
	Result	102.0...°C Acceptance the limits (100.0°C ± 4°C)
	☒ Passed ☐ Failed	
	Set point of 230°C	
	1. Set the oven temperature to 230°C	
	2. Allow the oven to stabilize This may take a few minutes	
	3. Observe the temperature on the calibrated thermometer's display and ,when the temperature stabilizer, record the reading	
Result	232.2...°C Acceptance the limits (230.0°C ± 5°C)	
☒ Passed ☐ Failed		
Note		

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

Inlet Leak Test

Inlet Leak Test	1. Zero the manometer and install it in place of the column in the inlet.	
	2. Cap the inlet's septum purge fitting with a 1/8 inch Swagelok cap.	
	3. Set the mode to constant pressure.	
	4. Establish a flow in the system (for example, inlet total flow of 50 ml/minute) and pressurize the inlet to 25 psi. This may take a few seconds. Note: The GC's keyboard/controller display may not read exactly 25.00 psi.	
	5. Wait 10 minutes for the system to equilibrate.	
	6. Turn off the inlet pressure but <u>do not</u> set the pressure value to zero.	
	7. From the GC's keyboard/controller display, record the initial pressure in the Results section.	
	8. Wait 5 minutes and then record the final pressure reading.	
	9. Calculate the difference and record it as the pressure drop.	
	10. Remove the septum purge vent cap and verify that the system begins to depressurize.	
	Result	0.37... psi Pressure Drop ($P_D = P_I - P_F$) ≤ 2.0
Note		

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

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Detector Flow Test

☐ Not install

Detector Flow Test

1. Turn off all detector flows except for the one being tested			
2. Set flow parameters specified in the evaluation parameter section			
3. Prepare the calibrated mass flow meter (gas selection, purge and zero).			
4. Connect the flowmeter to the detector exit			
5. Wait for the flowmeter's reading to stabilize			
6. Record the measured value in the results section			
7. Turn off the flow			
8. Repeat step 1 - 7 for each remaining detector flow			
9. After all flow have been measured, remove the cap from the detector base			
Result	<i>Hydrogen</i> (30 ± 3.0)	291.9...ml/min	☒ Passed ☐ Failed
	<i>Air</i> (400 ± 40.0)	385.8...ml/min	
	<i>Make up</i> (25 ± 2.5)	251.9...ml/min	

Chemical Evaluation Parameter

☐ Not install-ALS (Auto Liquid Sampler)

Chemical Evaluation Parameter	Description	Limit	Measure Values
	3. Precision peak area(RSD)**	≤ 5 %	4,022
	4. Precision Retention Time (RSD) **	≤ 0.50 %	0,019
	Result	☒ Passed ☐ Failed	
Note			

*For the measured value for the gas that is not used, enter N/A.If a sampler is used, the peak response is the average for all injection

**Enter N/A for the precision values if manual injection is used

*** The test gets a status of pass if all test results match the acceptance criteria

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

Date Printed: 09-Jun-2023 15:51:34

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Noise ,Wander and Drift Test

☐ Not install

Noise ,Wander and DriftTest	Description	Limit	Measure Values
	Base signal	$\leq 25 \text{ pA}$	20.6
	Noise	$\leq 0.10 \text{ pA}$	0.097
	Wander	$\leq 0.20 \text{ pA}$	0.027
	Drift	$\leq 2.50 \text{ pA /hours}$	0.202
	Result	☒ Passed ☐ Failed	
Note			

ETUNE.U and Tune Evaluation Test

☒ Not install

ATUNE.U and Tune Evaluation Test	1.From the Instrument & Control view, click Instrument > Edit MS Tune Parameters, select ATUNE.U, and then press OK The Edit Parameters screen is displayed.
	2. Select filament 1 and click the OK button.
	3. Select ATUNE.U and click the OK button to save the file.
	4. Perform the checkout tune for filament 1 (click Qualify > Checkout Tune).
	5. Perform the tune evaluation for the filament 1 (click Qualify > Tune Evaluation).
	6. Save this tune file as ATUNE_F1.U (click Instrument > Edit MS Tune Parameters, select ATUNE.U, press OK twice, and enter the new tune file name).
	7. Turn off the flow
	8. Repeat steps 1 – 7 for each retaining detector flow
	9. After all flows have been measured, remove the cap from the detector base.

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

Date Printed: 09-Jun-2023 15:51:34

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	Description	Limit	Measure values	Passed	Failed
	Ratio of mass 70 to mass 69	(0.5 – 1.6 %)		☐	☐
	Ratio of mass 220 to mass 219	(3.2 – 5.4 %)		☐	☐
	Ratio of mass 503 to mass 502	(7.9 – 12.3 %)		☐	☐
	Ratio of 219 to 69	>40%		☐	☐
	Ratio of 502 to 69	>2.4%		☐	☐
	Mass 69 precursor	≤ 3%		☐	☐
	Mass 219 precursor	≤ 6%		☐	☐
	Mass 502 precursor	≤ 12%		☐	☐
	Note				
The Electron Multiplier Voltage should be less than 2600 volt to accommodate the signal to noise test method that uses an EMV of relative + 400 Volts					

MSD
Testing for a leak in the system

☒ Not install

Testing for a leak in the system	Description	Limit	Measure values	Passed	Failed
	Position of mass 69	± 0.2		☐	☐
	Position of mass 219	± 0.2		☐	☐
	Position of mass 502	± 0.2		☐	☐
	Ratio of 18 to 69	< 20 %		☐	☐
	Ratio of 28 to 69	< 10%		☐	☐
	Electron Multiplier Voltage	≤ 2600		☐	☐

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

Date Printed: 09-Jun-2023 15:51:34

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Certification of System Qualification

OQ/PV Certification

Summary of Results		
All qualification tests and measurements were successfully completed.	Pass	☒
Some of the qualification tests or measurements required correction in order to successfully complete.*	Pass after correction	☐
Some of the qualification tests or measurements were not successfully completed.**	Fail	☐
<i>*Attach a Deviation Report that describes the corrective action that was taken. The Deviation Reports associated with this document are listed below. If a deviation is due to a process change (Product Support Bulletin, Service Note, and so on), attach a copy of the change notification.</i>		
<i>** Agilent Technologies recommends that the system be removed from service until it is repaired.</i>		

Deviations: Yes ___ No ___ Verified by (customer): _____

Comments: _____

ATTACHMENTS/SUPPORTING DOCUMENTS

The documents listed below are referred to but not included in the body of this protocol.

[illegible]

Notice

This document provides a protocol to verify and record instrument configuration and evidence of proper operation. It has been prepared from our interpretation of applicable regulations as well as industry best practices. The document is designed to provide an important component of a complete compliance package. Validation depends upon many factors and use of this protocol alone does not assure compliance. Agilent Technologies makes no promises or representations as to its sufficiency for any specific regulatory program.

Deviations: Yes No _____ Verified by (customer): _____

Comments:		P. 11 - f11
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Date of Performance	27/05/2026	Document Type	OQ
Engineer Name	Lâm Huỳnh Khang	Manufacturer / Brand	Agilent
Customer Company	CÔNG TY CỔ PHẦN ĐỒNG XANH	Equipment Type	GC
Customer Address	Lô G3-1 + G4+ MB1-4, khu công nghiệp Đức Hòa 1 - Hạnh Phúc, Xã Mỹ Hạnh, Tỉnh Tây Ninh, Việt Nam		

Module Configuration

Module	Serial Number
FID	CN16302004

Test Summary - OQ / Agilent / GC

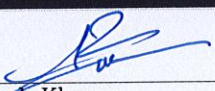
6 Test Parameters - OQ / Agilent / GC

Method / Test Parameter	Limit	Actual Value	Deviation	Judgment
Oven Temperature Accuracy				
Oven Temp 100 °C	100 ± 4 °C	102.000	2.00 °C	PASS
Oven Temp 230 °C	230 ± 5 °C	232.200	2.20 °C	PASS
Inlet Leak Test				
Pressure Drop (Pi – Pf)	≤ 2.0 psi	0.370	1.63 psi	PASS
Detector Flow Test				
Hydrogen Flow	30 ± 3.0 ml/min	29.900	0.10 ml/min	PASS
Air Flow	400 ± 40 ml/min	385.800	14.20 ml/min	PASS
Make-up Flow	25 ± 2.5 ml/min	25.900	0.90 ml/min	PASS
Chemical Evaluation — Injection Precision				
Peak Area (%RSD)	≤ 5.00 %	4.022	0.98%	PASS
Retention Time (%RSD)	≤ 0.50 %	0.019	0.48%	PASS

Method / Test Parameter	Limit	Actual Value	Deviation	Judgment
Noise, Wander & Drift (FID)				
Base Signal	$\leq 25 \text{ pA}$	20.600	4.40 pA	PASS
Noise	$\leq 0.10 \text{ pA}$	0.097	0.00 pA	PASS
Wander	$\leq 0.20 \text{ pA}$	0.027	0.17 pA	PASS
Drift	$\leq 2.50 \text{ pA/hour}$	0.202	0.20 pA/h	PASS

● PASS: 12/12 ● FAIL: 0/12 ● Pending: 0/12 **Overall: PASS**

Approval and Signatures

Performed By - VINTEK Engineer	Reviewed By - Customer Representative
 Lâm Huỳnh Khang Date: 27/05/2026	Name / Signature Date: _____

Detailed Component Results

Oven Temperature Accuracy

Purpose: This test verifies that the gas chromatograph oven maintains the set-point temperature within the manufacturer-specified accuracy tolerance at two reference temperatures (100 degrees C and 230 degrees C). Accurate oven temperature control is a fundamental requirement for reproducible retention times, consistent peak resolution, and reliable compound identification in gas chromatographic analyses. The test confirms proper thermocouple calibration and thermal regulation across the operational temperature range.

Set Point	Measured (°C)	Deviation (°C)	Limit	Result
100 °C	102.0	2.00	± 4.0 °C	PASS
230 °C	232.2	2.20	± 5.0 °C	PASS

Detailed Component Results

Inlet Leak Test

Purpose: This test evaluates the pneumatic integrity of the GC inlet system by pressurizing the inlet to a specified pressure and monitoring for pressure decay over a defined period. The measured pressure drop must not exceed the maximum allowable limit, confirming that all connections, septa, and sealing components are properly installed and free from leaks. A leak-free inlet is essential for accurate split ratio control, consistent sample introduction, and reliable quantitative gas chromatographic analysis.

Parameter	Value
Initial Pressure	25.00 psi
Final Pressure	24.63 psi
Pressure Drop	0.37 psi
Limit	≤ 2.0 psi
Result	PASS

Detailed Component Results

Chemical Evaluation - Injection Precision

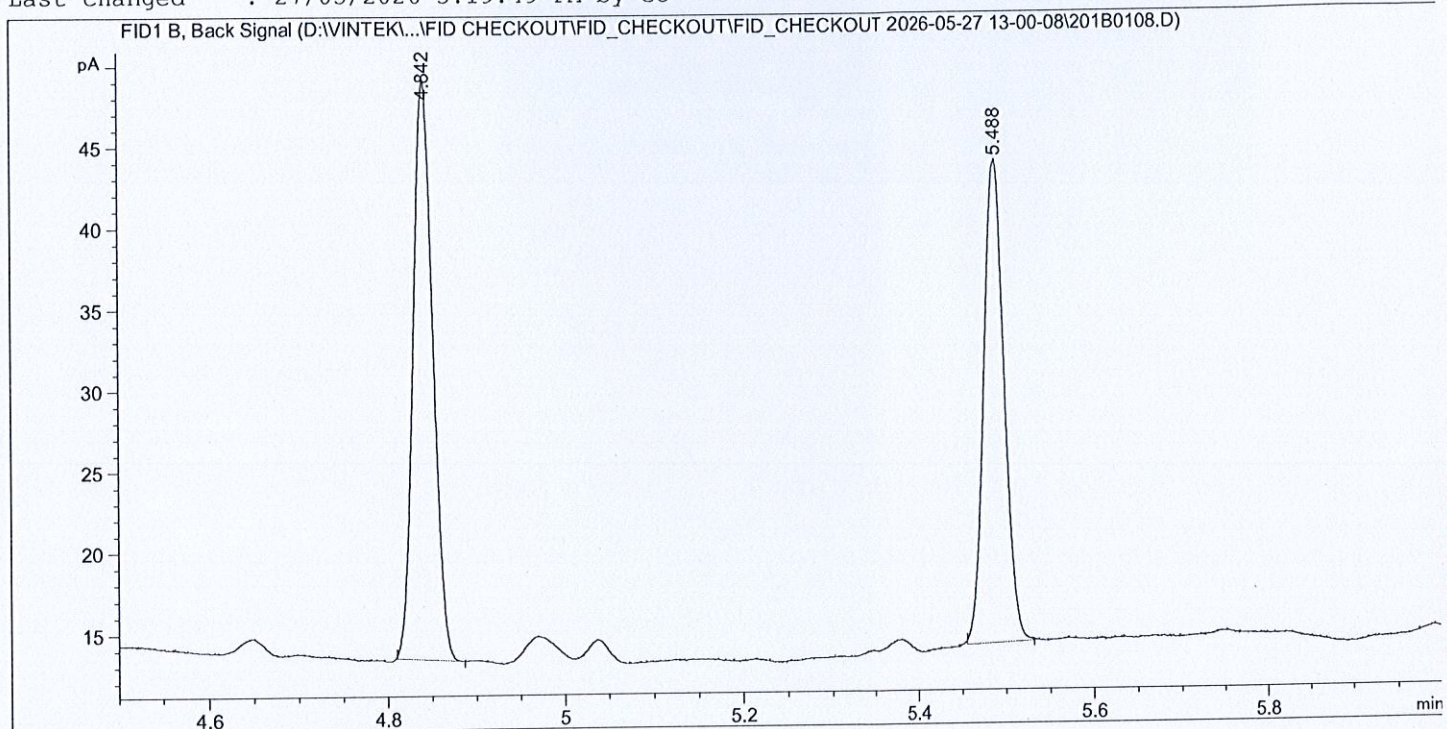
Purpose: This test evaluates the injection repeatability of the gas chromatograph by performing multiple consecutive injections of a reference standard and calculating the relative standard deviation (%RSD) for both peak area and retention time. The assessment confirms that the injection system delivers consistent sample volumes with minimal variability, which is a fundamental prerequisite for reliable quantitative gas chromatographic analysis and method validation.

Inj	Peak Area	Retention Time (min)
1	47.89	5.4880
2	43.86	5.4870
3	47.84	5.4850
4	43.86	5.4870
5	46.09	5.4860
6	44.92	5.4870
Average	45.74	5.4867
SD	1.8398	0.001033
%RSD	4.02%	0.02%

=====

Acq. Operator	: GC	Seq. Line	: 1
Acq. Instrument	: Instrument 1	Location	: Vial 201
Injection Date	: 27/05/2026 2:53:59 PM	Inj	: 8
		Inj Volume	: 1 µl
Acq. Method	: D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\FID_CHECKOUT.M		
Last changed	: 27/05/2026 2:06:56 PM by GC (modified after loading)		
Analysis Method	: D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\FID_CHECKOUT.M (Sequence Method)		
Last changed	: 27/05/2026 3:19:49 PM by GC		

=====



=====
Area Percent Report
=====

Sorted By : Signal
Multiplier : 1.0000
Dilution : 1.0000
Use Multiplier & Dilution Factor with ISTDs

Signal 1: FID1 B, Back Signal

Peak #	RetTime [min]	Type	Width [min]	Area [pA*s]	Height [pA]	Area %
1	4.842	BB	0.0245	55.13743	35.47405	53.51483
2	5.488	BB	0.0252	47.89464	29.69400	46.48517

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Sample Name: FID_CHECKOUT

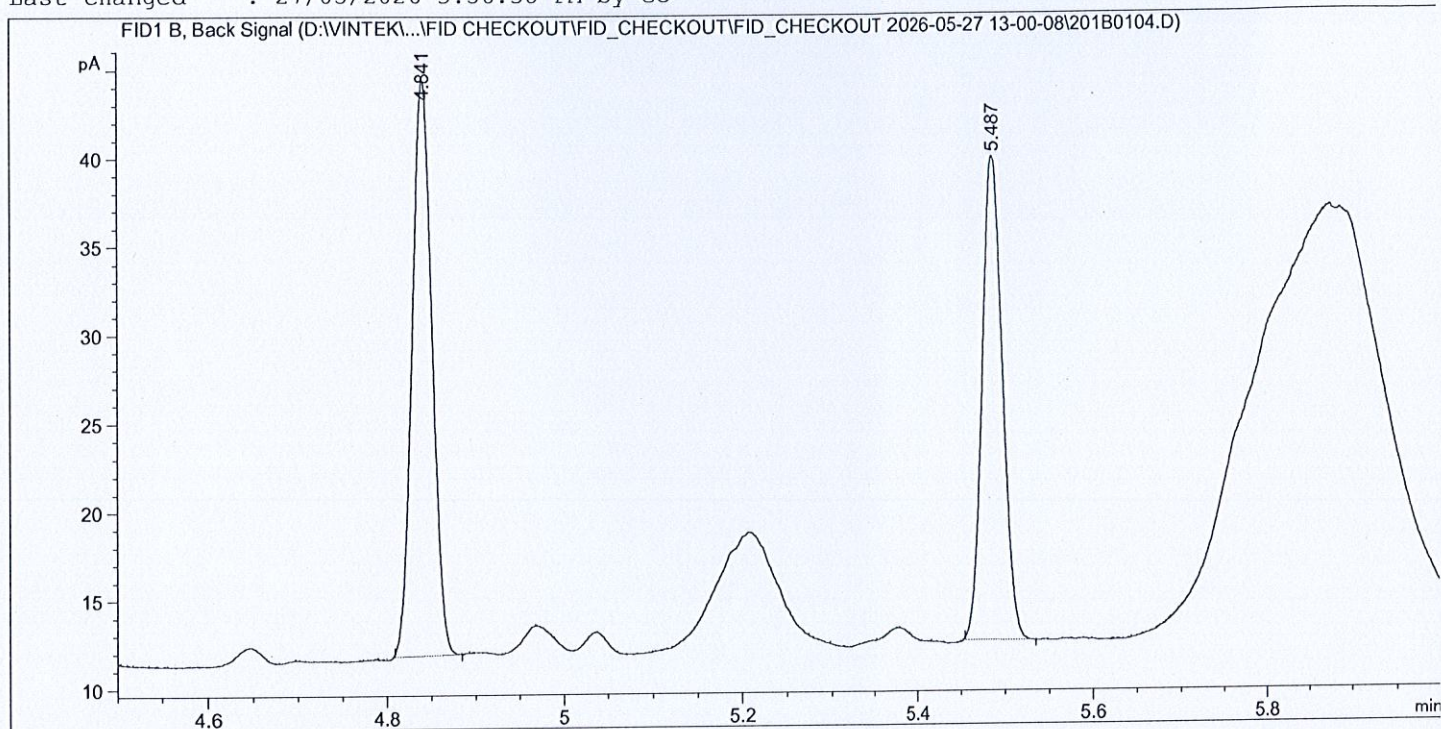
Totals : 103.03207 65.16805

=====
*** End of Report ***

=====

Acq. Operator	: GC	Seq. Line	: 1
Acq. Instrument	: Instrument 1	Location	: Vial 201
Injection Date	: 27/05/2026 1:47:30 PM	Inj	: 4
		Inj Volume	: 1 µl
Acq. Method	: D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\FID_CHECKOUT.M		
Last changed	: 12/07/2024 4:24:48 PM by GC		
Analysis Method	: D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\FID_CHECKOUT.M (Sequence Method)		
Last changed	: 27/05/2026 3:56:38 PM by GC		

=====



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Area Percent Report

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Sorted By : Signal
Multiplier : 1.0000
Dilution : 1.0000
Use Multiplier & Dilution Factor with ISTDs

Signal 1: FID1 B, Back Signal

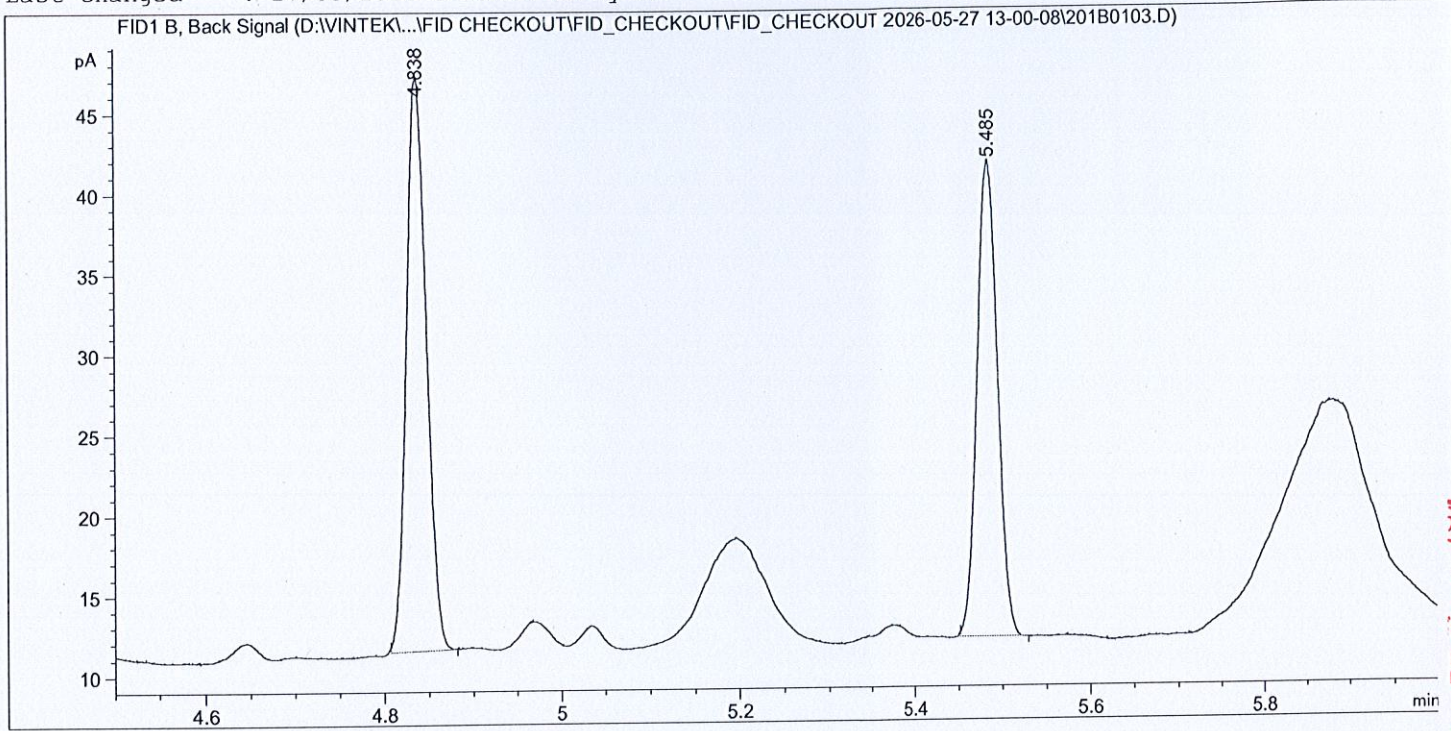
Peak #	RetTime [min]	Type	Width [min]	Area [pA*s]	Height [pA]	Area %
1	4.841	BB	0.0247	50.82394	32.44595	53.68006
2	5.487	BB	0.0259	43.85543	27.33765	46.31994

Totals : 94.67937 59.78360

Data File D:\VINTEK\...D CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\201B0104.D
Sample Name: FID_CHECKOUT

=====
*** End of Report ***


```
=====
Acq. Operator   : GC                      Seq. Line :    1
Acq. Instrument : Instrument 1             Location  : Vial 201
Injection Date  : 27/05/2026 1:33:09 PM   Inj       :    3
                                           Inj Volume: 1 µl
Acq. Method     : D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13
                  -00-08\FID_CHECKOUT.M
Last changed    : 12/07/2024 4:24:48 PM by GC
Analysis Method : D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13
                  -00-08\FID_CHECKOUT.M (Sequence Method)
Last changed    : 27/05/2026 3:19:49 PM by GC
=====
```



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=====
                          Area Percent Report
=====
```

Sorted By : Signal
Multiplier : 1.0000
Dilution : 1.0000
Use Multiplier & Dilution Factor with ISTDs

Signal 1: FID1 B, Back Signal

Peak #	RetTime [min]	Type	Width [min]	Area [pA*s]	Height [pA]	Area %
1	4.838	BB	0.0247	55.92734	35.66986	53.89924
2	5.485	BB	0.0253	47.83543	29.47169	46.10076

Totals : 103.76277 65.14155

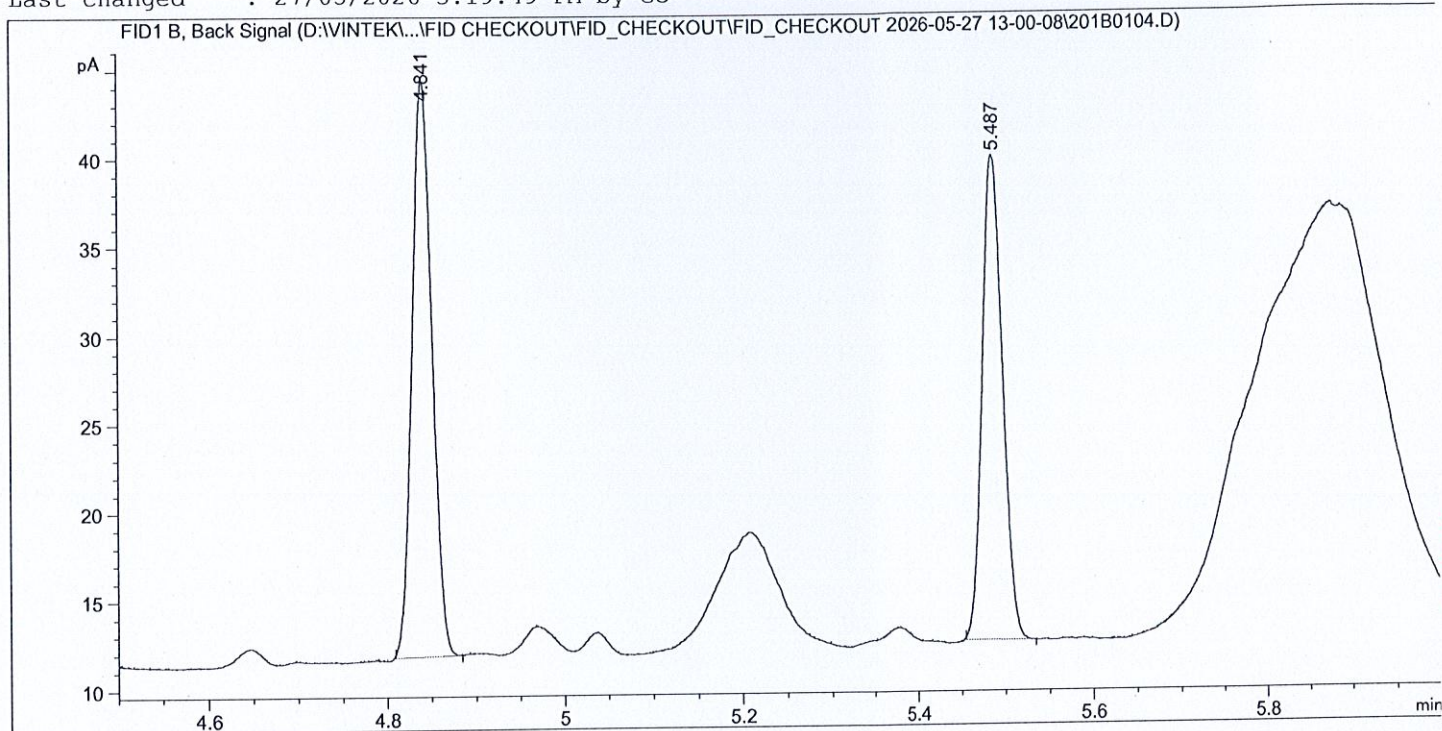
Data File D:\VINTEK\...D CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\201B0103.D
Sample Name: FID_CHECKOUT

=====
*** End of Report ***

=====

Acq. Operator	: GC	Seq. Line	: 1
Acq. Instrument	: Instrument 1	Location	: Vial 201
Injection Date	: 27/05/2026 1:47:30 PM	Inj	: 4
		Inj Volume	: 1 µl
Acq. Method	: D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\FID_CHECKOUT.M		
Last changed	: 12/07/2024 4:24:48 PM by GC		
Analysis Method	: D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\FID_CHECKOUT.M (Sequence Method)		
Last changed	: 27/05/2026 3:19:49 PM by GC		

=====



=====
Area Percent Report
=====

Sorted By : Signal
Multiplier : 1.0000
Dilution : 1.0000
Use Multiplier & Dilution Factor with ISTDs

Signal 1: FID1 B, Back Signal

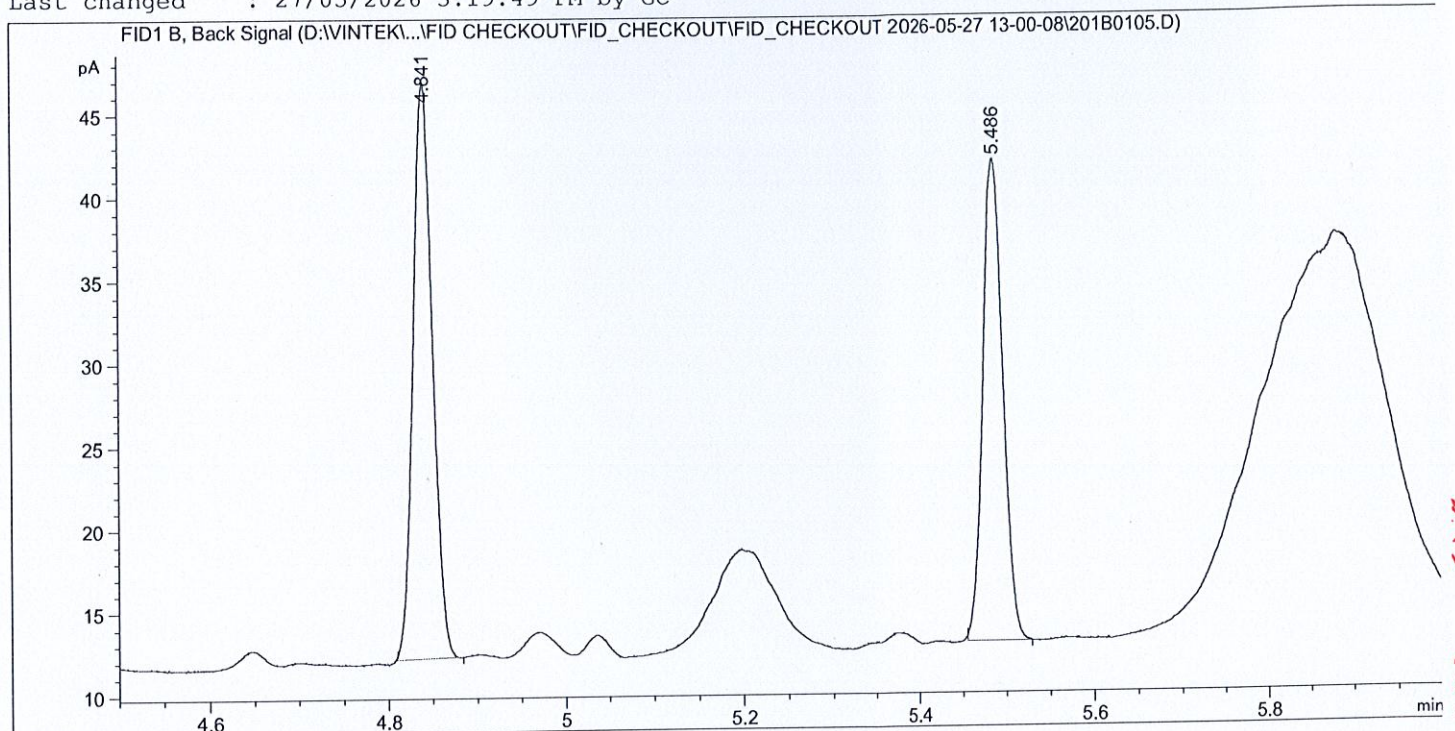
Peak #	RetTime [min]	Type	Width [min]	Area [pA*s]	Height [pA]	Area %
1	4.841	BB	0.0247	50.82394	32.44595	53.68006
2	5.487	BB	0.0259	43.85543	27.33765	46.31994

Totals : 94.67937 59.78360

Data File D:\VINTEK\...D CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\201B0104.D
Sample Name: FID_CHECKOUT

=====
*** End of Report ***


```
=====
Acq. Operator   : GC                      Seq. Line :    1
Acq. Instrument : Instrument 1             Location  : Vial 201
Injection Date  : 27/05/2026 2:01:51 PM   Inj       :    5
                                           Inj Volume: 1 µl
Acq. Method     : D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13
                                           -00-08\FID_CHECKOUT.M
Last changed    : 27/05/2026 2:06:56 PM by GC
                                           (modified after loading)
Analysis Method : D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13
                                           -00-08\FID_CHECKOUT.M (Sequence Method)
Last changed    : 27/05/2026 3:19:49 PM by GC
=====
```



```
=====
Area Percent Report
=====
```

Sorted By : Signal
Multiplier : 1.0000
Dilution : 1.0000
Use Multiplier & Dilution Factor with ISTDs

Signal 1: FID1 B, Back Signal

Peak #	RetTime [min]	Type	Width [min]	Area [pA*s]	Height [pA]	Area %
1	4.841	BB	0.0243	53.13810	34.64181	53.55071
2	5.486	BB	0.0242	46.09139	28.96326	46.44929

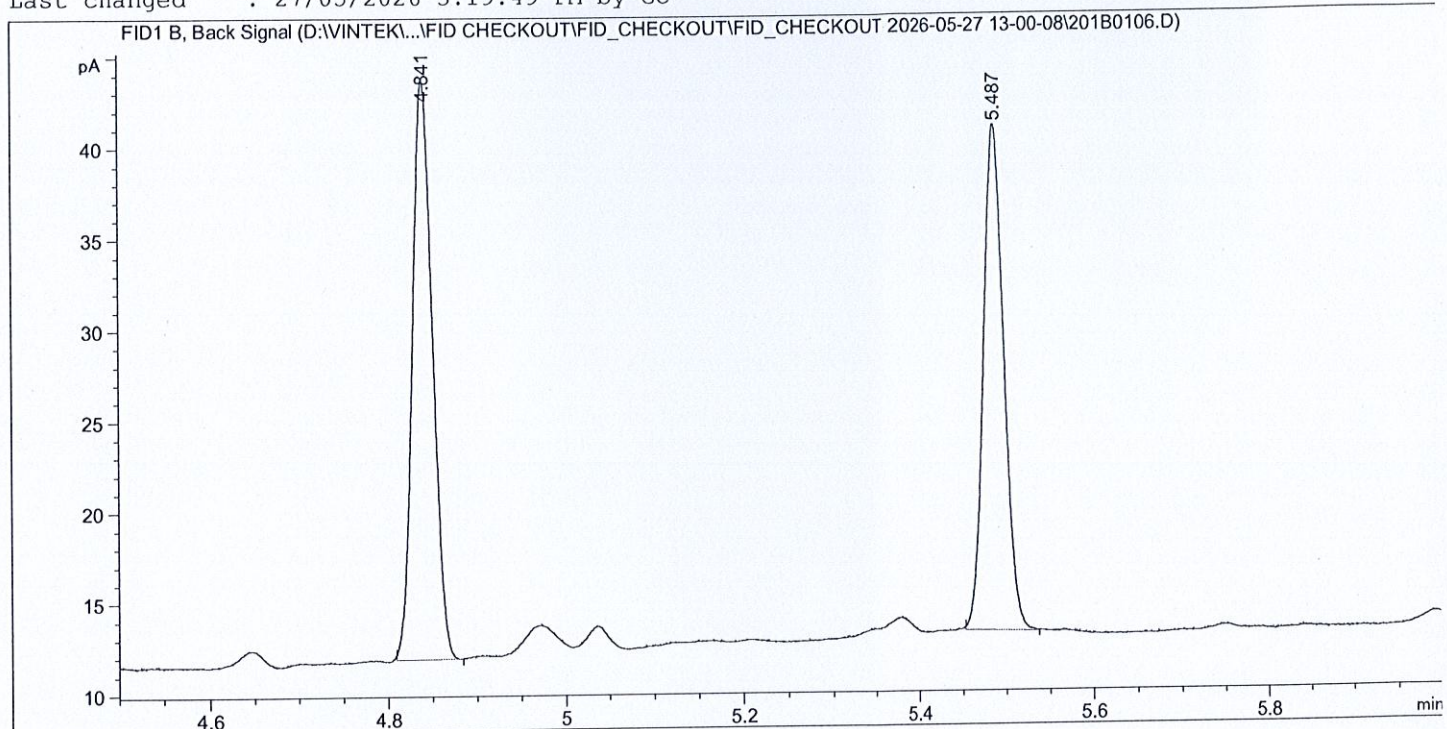
Data File D:\VINTEK\...D CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\201B0105.D
Sample Name: FID_CHECKOUT

Totals : 99.22949 63.60507

=====
*** End of Report ***


```
=====
Acq. Operator   : GC                               Seq. Line :    1
Acq. Instrument : Instrument 1                     Location  : Vial 201
Injection Date  : 27/05/2026 2:19:15 PM           Inj       :    6
                                                Inj Volume: 1 µl

Acq. Method     : D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13
                  -00-08\FID_CHECKOUT.M
Last changed    : 27/05/2026 2:06:56 PM by GC
                  (modified after loading)
Analysis Method : D:\VINTEK\27.05.2026\DATA\FID_CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13
                  -00-08\FID_CHECKOUT.M (Sequence Method)
Last changed    : 27/05/2026 3:19:49 PM by GC
=====
```



```
=====
                        Area Percent Report
=====
```

Sorted By : Signal
Multiplier : 1.0000
Dilution : 1.0000
Use Multiplier & Dilution Factor with ISTDs

Signal 1: FID1 B, Back Signal

Peak #	RetTime [min]	Type	Width [min]	Area [pA*s]	Height [pA]	Area %
1	4.841	BB	0.0250	50.38316	31.66580	52.86348
2	5.487	BB	0.0261	44.92490	27.69677	47.13652

Data File D:\VINTEK\...D CHECKOUT\FID_CHECKOUT\FID_CHECKOUT 2026-05-27 13-00-08\201B0106.D
Sample Name: FID_CHECKOUT

Totals : 95.30806 59.36256

=====
*** End of Report ***

Detailed Component Results

FID - Noise, Wander & Drift

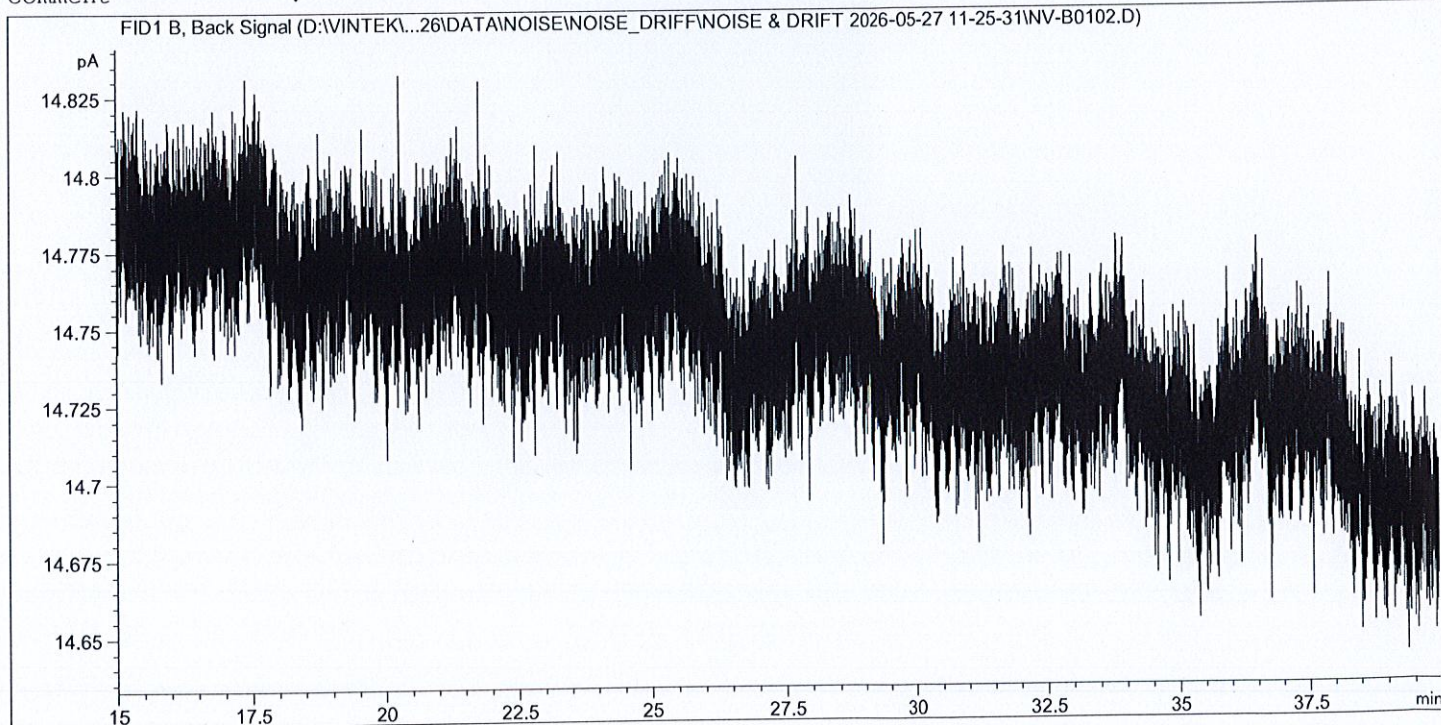
Purpose: This test evaluates the baseline performance characteristics of the Flame Ionization Detector (FID) by quantifying the short-term noise, intermediate-term wander, and long-term drift under stable operating conditions. These parameters collectively define the detection limit capability and long-term stability of the FID system. Compliance with specified limits confirms that the detector maintains adequate sensitivity and baseline stability for reliable trace-level quantitative analysis and extended analytical sequences.

Parameter	Measured	Limit	Result
Base Signal	20.6 pA	≤ 25 pA	PASS
Noise	0.0970 pA	≤ 0.10 pA	PASS
Wander	0.0272 pA	≤ 0.20 pA	PASS
Drift	0.2022 pA/h	$\leq 2.50 $ pA/h	PASS


```
=====
Acq. Operator   : GC                      Seq. Line :    1
Acq. Instrument : Instrument 1             Location  :    -
Injection Date  : 27/05/2026 12:12:59 PM   Inj       :    2
                                           Inj Volume : No inj
Acq. Method     : D:\VINTEK\27.05.2026\DATA\NOISE\NOISE_DRIFT\NOISE & DRIFT 2026-05-27 11-25-31
                                           \NOISE DRIFT.M
Last changed    : 27/05/2026 11:26:46 AM by GC
                                           (modified after loading)
Analysis Method : D:\VINTEK\27.05.2026\DATA\NOISE\NOISE_DRIFT\NOISE & DRIFT 2026-05-27 11-25-31
                                           \NOISE DRIFT.M (Sequence Method)
Last changed    : 27/05/2026 12:56:41 PM by GC
Method Info     : Noise Drift
=====
```

```
=====
Column(s)
=====
```

Column Description : HP-5 5% Phenyl Methyl Siloxane
Inventory# : 3522.58410
Model# : 19091J-413 Manufacturer: Agilent
Diameter : 320.00 µm Length : 30.0 m
Film thickness : 0.25 µm Void time : 1.309 min
Injections : 37201
Maximum Temperature: 325.0 °C
Comment :



```
=====
Area Percent Report with Performance and Noise
=====
```


Sample Name: Noise & Drift

Multiplier : 1.0000
Dilution : 1.0000
Use Multiplier & Dilution Factor with ISTDs

Signal 1: FID1 B, Back Signal

Noise determination:

Time range		Noise	Noise	Noise		
from	to	(6*SD)	(PtoP)	(ASTM)	Wander	Drift
[min]	[min]	[pA]	[pA]	[pA]	[pA]	[pA/h]
15.000	40.000	8.573e-2	9.291e-2	9.696e-2	2.715e-2	-2.022e-1

*** End of Report ***



ỦY BAN TIÊU CHUẨN ĐO LƯỜNG CHẤT LƯỢNG QUỐC GIA
TRUNG TÂM KỸ THUẬT TIÊU CHUẨN ĐO LƯỜNG CHẤT LƯỢNG 3
QUALITY ASSURANCE & TESTING CENTER 3



Head Office: 49 Pasteur, Dist. 1, HCMC, Vietnam Tel: (84-28) 3829 4274 Fax: (84-28) 3829 3012 E-mail: info@quatest3.com.vn Website: www.quatest3.com.vn
Testing Complex: 6 No.7, road No.1, Bien Hoa 1 Z, Dong Nai, Vietnam 6 C5 lot, K1 road, Cat Lai Z, Thu Duc City, HCMC, Vietnam 6 64 Le Hong Phong, Dist.5, HCMC, Vietnam

KT3-00981ANH5

GIẤY CHỨNG NHẬN HIỆU CHUẨN
CERTIFICATE OF CALIBRATION

Số giấy chứng nhận đăng ký cung cấp dịch vụ/ Service License N°: ĐK 03

03/06/2025
Page : 01/02

1. Phương tiện đo:
Object

ĐỒNG HỒ ĐO NHIỆT ĐỘ HIỆN SỐ
DIGITAL THERMOMETER

2. Nơi sản xuất
Manufacturer

FLUKE

3. Kiểu/ Type:

17B+

SN: 46176868WS

ID: N/A

4. Đặc trưng kỹ thuật:
Specification

+ Phạm vi đo/ Range:

- 55 °C đến/ to 400 °C

+ Độ phân giải/ Resolution:

0,1 °C

+ Cặp nhiệt điện/ Thermocouple: Type K

5. Khách hàng:
Customer

CÔNG TY TNHH VINTEK

280/130 Bùi Hữu Nghĩa, P2, Bình Thạnh, TP Hồ Chí Minh

6. Nơi hiệu chuẩn:
Place of Calibration

TRUNG TÂM KỸ THUẬT 3/ QUATEST 3

7 Đường 1, KCN Biên Hòa 1, Đồng Nai

7. Phương pháp hiệu chuẩn:
Method of Calibration

QTHC/KT3 17: 2022 Nhiệt kế hiện số & tương tự - Quy trình hiệu chuẩn
Digital & Analog Thermometers - Calibration Procedure

8. Chuẩn sử dụng/Standards Used:

ID	Description	Cal. Date	Due Date	Traceable to
NH1861	Temperature Indicator	02/2025	02/2026	VMI-VIETNAM
NH2007	PRT, Pt 100	02/2025	02/2026	VMI-VIETNAM
NH2080	Calibration Bath	12/2024	12/2025	VMI-VIETNAM
NH1790; NH2081	Calibration Bath	12/2024	12/2025	VMI-VIETNAM

9. Môi trường hiệu chuẩn/ Calibration Environment:

[23 ± 5] °C

[50 ± 20] %RH

10. Hiệu chỉnh phương tiện đo/ Adjustment:

Không/ No

11. Ngày hiệu chuẩn/ Date of Calibration:

02/06/2025

12. Tem hiệu chuẩn/ Calibration Label:

KT3-00981ANH5

13. Ngày hiệu chuẩn lại theo yêu cầu của khách hàng:

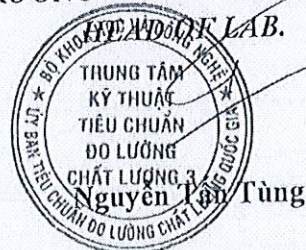
02/06/2026

Recalibration Date as Request of Customer

TRƯỞNG PDL NHIỆT
HEAD OF TEMPERATURE MEAS. LAB.

Trần Lê Phương

TL. GIÁM ĐỐC/ PP. DIRECTOR
TRƯỞNG PHÒNG THÍ NGHIỆM



1. Giấy chứng nhận hiệu chuẩn này thể hiện việc liên kết chuẩn đến chuẩn quốc gia, với đơn vị đo tuân thủ theo hệ đơn vị đo quốc tế SI.
This calibration certificate documents the traceability to national standards, which realize the units of measurement according to the International System of Units.

2. Độ không đảm bảo đo mở rộng được tính từ độ không đảm bảo đo chuẩn nhân với hệ số phủ k = 2, phân bố chuẩn tương ứng với 95 % độ tin cậy.
The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.

3. Không được trích sao một phần giấy chứng nhận hiệu chuẩn này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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4. Mọi thắc mắc về giấy chứng nhận hiệu chuẩn, khách hàng liên hệ theo địa chỉ dh.cs@quatest3.com.vn để biết thêm thông tin.
Please contact QUATEST 3 at the email addresses dh.cs@quatest3.com.vn for further information about certificate of calibration.

5. Phòng Thí nghiệm đã được Văn phòng Công nhận Chất lượng (BoA)-Việt Nam công nhận phù hợp theo ISO/IEC 17025:2017 (Vilas 036). Các nội dung có dấu (*) là chưa được công nhận.
The Testing Lab is accredited as conforming to ISO/IEC 17025:2017 by Bureau of Accreditation - Viet Nam (Vilas 036). The characteristics marked with (*) are not accredited yet.





ỦY BAN TIÊU CHUẨN ĐO LƯỜNG CHẤT LƯỢNG QUỐC GIA
TRUNG TÂM KỸ THUẬT TIÊU CHUẨN ĐO LƯỜNG CHẤT LƯỢNG 3
QUALITY ASSURANCE & TESTING CENTER 3

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Testing Complex: 9 No.7, road No.1, Bien Hoa 1 Z, Dong Nai, Vietnam 9 C5 lot, K1 road, Cat Lai Z, Thu Duc City, HCMC, Vietnam



KT3-00981ANH5

GIẤY CHỨNG NHẬN HIỆU CHUẨN
CERTIFICATE OF CALIBRATION

Số giấy chứng nhận đăng ký cung cấp dịch vụ/ Service License N°: ĐK 03



06/2025
Date: 02/02

14. Kết quả hiệu chuẩn/Results of calibration

Đầu dò Probe	Giá trị chỉ thị, °C Indication Value	Giá trị chuẩn, °C Reference Value	Số hiệu chỉnh, °C Correction	Độ KĐBD, °C Uncertainty
Type K [2]	36,2	37,00	+ 0,8	0,2
	38,9	40,00	+ 1,1	0,2
	58,9	60,00	+ 1,1	0,2
	99,1	100,00	+ 0,9	0,2
	229,3	230,00	+ 0,7	0,2

15. Thông tin khác/ Other Informations

a. Các giá trị có đơn vị đo không thuộc hệ SI, được chuyển đổi từ hệ SI theo các bảng trong tài liệu
NĐ 86/2012/NĐ-CP; BIPM SI Brochure.

All non-SI values were converted from SI units via conversion factors in above documents.

b. Kết quả hiệu chuẩn chỉ có giá trị tương ứng với điều kiện theo phương pháp hiệu chuẩn nêu ở Mục 7.
Calibration results are valid with respect to the procedure conditions as description at Item 7. only.

c. Các kết quả hiệu chuẩn được thực hiện với mười lần đo để tính giá trị trung bình và hiệu chỉnh.
All calibration results are based on times of measurement, from which the average and correction are calculated.

d. Các giá trị của đại lượng được hiệu chuẩn theo yêu cầu của khách hàng.
The quantity values are calibrated as request of customer.

e. Khả năng đọc trên phương tiện đo/ Readability:

0,1 °C
200 mm

f. Mức nhúng/ Immersion:

- Giấy chứng nhận hiệu chuẩn này thể hiện việc liên kết chuẩn đến chuẩn quốc gia, với đơn vị đo tuân thủ theo hệ đơn vị đo quốc tế SI.
This calibration certificate documents the traceability to national standards, which realize the units of measurement according to the International System of Units.
- Độ không đảm bảo đo mở rộng được tính từ độ không đảm bảo đo chuẩn nhân với hệ số phủ k = 2, phần bố chuẩn tương ứng với 95 % độ tin cậy.
The reported expanded uncertainty of measurement is stated as the standard uncertainty multiplied by the coverage factor k = 2, at 95 % confidence level.
- Không được trích sao một phần giấy chứng nhận hiệu chuẩn này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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- Mọi thắc mắc về giấy chứng nhận hiệu chuẩn, khách hàng liên hệ theo địa chỉ dh.cs@quatest3.com.vn để biết thêm thông tin.
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- Phòng Thí nghiệm đã được Văn phòng Công nhận Chất lượng (BQA)-Việt Nam công nhận phù hợp theo ISO/IEC 17025:2017 (Vilas 036). Các nội dung có dấu (*) là chưa được công nhận.
The Testing Lab is accredited as conforming to ISO/IEC 17025:2017 by Bureau of Accreditation - Viet Nam (Vilas 036). The characteristics marked with (*) are not accredited yet.



Agilent Technologies



Certificate of Analysis

FID MDL Sample

Agilent Part
Number: 5188-5372

Sample Lot
Number: CG-3690

This Certified Reference Material (CRM) was manufactured and verified in accordance with ULTRA Scientific's ISO 9001 registered quality system, and the analyte concentrations were verified by our ISO 17025 accredited laboratory. The certified value, and uncertainty value, for each analyte was determined gravimetrically.

Concentrations:

n-tridecane	2.37 mg/L ($\pm 0.01\%$)
n-tetradecane	2.37 mg/L ($\pm 0.01\%$)
n-pentadecane	23.6 mg/L ($\pm 0.1\%$)
n-hexadecane	23.7 mg/L ($\pm 0.1\%$)

Solvent: isooctane

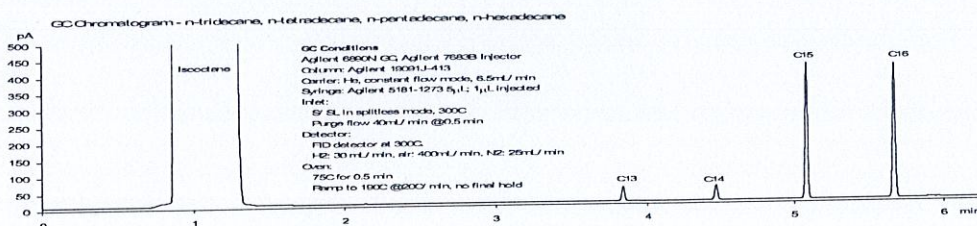
ULTRA Scientific uses calibrated Class A glassware and clean bottles in the manufacture of ULTRA standards. Balances used in the manufacture of this standard are calibrated with weights traceable to NIST in compliance with ANSI/NCSL Z-540-1 and ISO 9001.

Purities:

n-tridecane	99%
n-tetradecane	99%
n-pentadecane	99%
n-hexadecane	99%
isooctane	99%

Typical Analytical Spectrum or Chromatography

GC Chromatography – n-tridecane, n-tetradecane, n-pentadecane and n-hexadecane in isooctane



Date of release: 28 October 2024
Date of expiration: 30 November 2027

William J. Leary
Quality Assurance Manager



Certificate of Completion

Learner Name:

KHANG LAM HUYNH

Title Of Course:

AN-ASP/CE/CSE-GC-1-018-P:8890 and 7890 GC Support SPIFPM, Troubleshooting and Repair

Completion Date:

July 12, 2024

Certified By Company:

Learning at Agilent



All Service and Support training certificates have the following specific limitations.

A certificate for Service and Support training is only valid while employed by Agilent Technologies or while working as an Agilent-authorized service provider, through which the service employee has ongoing access to Agilent's: Safety Alerts, Service Notes, internal technical updates, update training, current documentation, technical support, current parts, and parts updates. Completion of training alone, without being employed by Agilent Technologies, does not qualify an individual to safely install, service or maintain Agilent products.